PRE-REGISTRATION — final round robin (human vs model demonstrations)

Drafted 2026-08-23 on the monitoring box from the 08-22/23 audits (paper/AUDIT_sources_2026-08-23.md, paper/CRITIQUE_design_final_rr_2026-08-23.md, PAPER_PLAN decision log 2026-08-23). Status: **DRAFT for user sign-off; nothing below is launched.** Once signed, this file is frozen for the block; amendments are appended with a date, never edited in place.

User decisions already taken (08-23): one decision clock (repeat 4) for every learner and every tape; retrain everything; run **sparse AND dense** for the RL and WM learners; keep dv3 at a longer budget but **queue it last**; implement every confound protection in CRITIQUE §4 except the TD-target clamp (kept out of the matrix; optional ablation).

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0. The one-sentence design

Three demonstration sources (dH human, dDP diffusion-policy teacher, dR2D r2dreamer teacher), **recorded by ONE recorder through the learners' own MDP** (FullTaskEnv, pick scope, delta_joint, cap 0.025, leash 0.125, action_repeat 4, tip rule, 1200 sim-step horizon), matched in N and in IC coverage, success-only in the primary matrix; consumed by four learners (DP, RLPD, r2dreamer, dv3) at that clock, evaluated by ONE harness (fresh process per episode, 1200 sim steps, one shared IC file, K=5 archived checkpoints, selection/confirmation split); RL/WM learners run sparse and dense.

Notation unchanged: d{source}_{learner}[-dense], e.g. dR2D_RLPD-dense_s3.

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1. Matrix

```
learner sources seeds reward budget (UNIT) secondary arms
----------------- ------------ ----- ----------------
-----------------------------------------------------------
-----------------------------------------------------
DP dH, dDP, dR2D 5 — 100k grad steps AND 300k (epoch-matched view; see §6) dHunpruned_DP (preprocessing control)
×3
RLPD dH, dDP, dR2D 10 sparse and dense 100k DECISIONS (= 400k sim steps) dDP+fails, dR2D+DPfails ×6 (sparse; N1
mechanism arms)
r2dreamer dH, dDP, dR2D 6 sparse and dense 3M SIM steps (750k decisions) —
dv3 (queued LAST) dH, dDP, dR2D 4 sparse and dense 1M SIM steps (250k decisions = 62.5k updates at train_ratio —
256)
```

- Seed indices shared across sources within a learner (paired blocks); new seed values never used before for that learner (disjoint from RUN_REGISTRY history).
- Queue order: pilots → DP → RLPD → r2dreamer → RLPD secondary → dv3.
- Job count: DP 15(+3) · RLPD 60(+12) · r2d 36 · dv3 24 = 150 jobs; GPU-h ≈

DP 54 + RLPD 216 + r2d 576 + dv3 ~400 ≈ 1250 (r2d and dv3 dominate; dv3 last by construction). Cut order if forced: dv3 seeds 4→3; r2d 6→5 (not lower); RLPD secondary 6→4; DP 300k view.

2. Held fixed (all learners, all sources, all arms)

```
knob value where enforced
------------------ -------------------------------------------------------------
---------------------------------------------------------
decision clock action_repeat 4 env ctor arg + tape stamp + sidecar + registry knob
action space delta_joint, delta_ref=target, cap 0.025 rad/sim-step, leash FullTaskEnv; DP emits absolute window-
end targets and is
0.125 executed through the same integrator at eval (hold-4)
training horizon 1200 sim steps = 300 decisions env max_steps; r2d/dv3 time_limit 1200 (pilot-gated)
terminal hardened pick (+1; ×100 for WMs), tip rule (tilt>60° ∧ FullTaskEnv online; terminal_from_tape offline
(§4.3)
grip<0.3 → terminate, r=0), timeout = truncation
dense variant r + γφ(s') − φ(s), φ = −2‖eef−can‖, φ(terminal)=0, γ = the full_env.step; encoders
learner's discount (RLPD 0.998, dv3 0.997, r2d = its config
value, VERIFIED), demo transitions relabelled with the same
potential from recorded eef_pos
demo preprocessing identical for all sources: recorded-as-executed tapes (§4), recorder + set builder
success-only primary, fails only in the named secondary
arms, tape ends where the env terminated
eval §5 cluster/eval_sweep.sh + eval_ics.json
code one commit hash for the block; a fix restarts that learner's registry
block
```

3. Demo-set protocol

3.1 Sources

- **dH**: the 66 IL-usable success demos (PAPER_PLAN §3). Leading-idle pruning applied ONCE (make_dp_pruned rule) before re-recording — the pruned set is the primary for all learners; the unpruned set is the dHunpruned_DP control only. Recorded by the human-waypoint adapter (§4.2) = closed-loop follower of the demo's own command stream at repeat 4 in the target-ref delta env. Kept iff it re-earns the hardened pick (replay gate). Report survivors /66.
- **dDP**: teacher = the DP trained on dH in THIS block's DP-r4 pilot (the median in-dist seed of the new dH_DP row — rule fixed now, not the best seed). Recorded by the dp adapter (hold-4, absolute targets → delta via the env integrator).
- **dR2D**: teacher = r2dreamer CHAMPION_1576820.pt (fixed). Recorded by the r2d adapter.
- Both teachers: `--mode sample`, attempts ≤3 per IC, IC list = the same success uids cycled, sim cap 1200, `--verify`, random-teacher negative control (must keep ≈0), ALL fails kept to `<set>_fails/`, manifest records teacher success rate on the harvest ICs.
- Teacher circularity and competence are disclosure (both teachers descend from dH).

3.2 Matching

N = min(dH survivors, dDP successes, dR2D successes), capped at 66; larger sets are subsampled uniformly (fixed seed) **stratified by ic_uid** so the three sets cover the same uids with the same multiplicity. Frames are NOT matched (intrinsic; reported). Per-set census (analysis/characterize_demo_sets.py, incl. the stride table) is published BEFORE launch; the per-episode list is sha256'd and every sbatch asserts the sha.

3.3 Fails arms (RLPD sparse only)

dDP+fails = the N dDP successes + dDP fails (tip-terminated by the recorder, guard on), fail share capped at the human no-pick share (~30% of episodes); dR2D+DPfails = the same fails injected into dR2D. Pre-registered predictions in PAPER_NOTES N1 carry over (ignition of dDP success-only ≥ dH; adding DP fails to dR2D lowers it by ≥0.15).

4. THE TAPE CONTRACT (v1) — what the recorder writes, what every consumer reads

One recorder, `baselines/record_demos.py` (NEW), teacher adapters `human | dp | r2d | random`, env = `FullTaskEnv(scope='pick', action_mode='delta_joint', delta_ref='target', delta_cap=0.025, delta_leash=0.125, action_repeat=4, max_steps=1200, camera_rig=True)` (CPU). One npz per episode, filename = rollout index ≥100000 (never a human uid); keys:

```
key shape meaning
----------------------- ------------------- ------------------------------------------------------------
-------------------------------------------------- ----------------------------------------------
states (n,17) f32 obs BEFORE each decision (joints 6, grip, grip effort, can
xyz, can quat wxyz, goal xy)
final_state (17,) f32 obs after the last decision
actions_delta (n,7) f32 the normalized [-1,1]^7 decision the env EXECUTED (arm
deltas, grip) — the learners' action space, exact
actions (n,7) f32 absolute command at the window END ([6 joint targets rad,
grip 0..1]) — DP/jact and legacy consumers
rewards (n,) f32 env reward per decision (sparse; shaping is NOT baked in)
terminated (n,) bool env terminated flag per decision
truncated (n,) bool env truncated flag (timeout)
picked (n,) bool env hardened-pick flag per decision
tipped (n,) bool env tip flag per decision
eef_pos (n+1,3) f32 tool position before each decision + final (for demo shaping
/ offline hardened predicate)
images (n+1,64,64,6) u8 top ++ wrist RGB before each decision + final (camera rig;
required for WM sets)
sim_states, sim_actions (4n,17), (4n,7) f32 per-sim-step sub-tape (states before each sim step; absolute
command each sim step) — lets any stride ≤4 be derived
later; optional but recorded by default
scalars uid (rollout idx), ic_uid, label ('success' 'fail'), stage, teacher, teacher_ckpt, act_mode (mode
sample), action_repeat=4, delta_cap, delta_leash,
delta_ref='target', pick_z, n, git_sha, env_class,
recorder='record_demos.py v1', contract='v1'
```

Rules: the tape ENDS at the env's terminal (terminated=True on the last row) or at the cap (truncated=True on the last row) — never later. `label`='success' iff the last row has picked=True. The recorder asserts `terminated|truncated` on exactly the last row. Directory gets `manifest.json` (teacher, ICs, attempts, kept/dropped, negctl result, content sha256, git sha, contract version).

4.1 Consumers (no consumer re-derives anything the tape already carries)

- **RLPD** `--demo-format native`: transitions (states[t], actions_delta[t], rewards[t]

(+shaping from eef_pos when dense), next = states[t+1] or final_state, done = terminated[t]); truncation bootstraps. No delta re-encoding, no relabel predicate.

- **DP**: lerobot dataset at fps 30/4 = 7.5 from (states, actions) per decision; eval and harvest execute DP through the same hold-4 integrator.
- **WM** (r2d/dv3): dreamer npz written DIRECTLY from the tape (`to_dreamer_native.py`, NEW): image[t], action[t] (backward-shifted, zeros at 0), reward ×100 at the terminal decision, is_terminal at terminated, is_last at the end, discount = 1−is_terminal; no grant slack; repeat.json stamp `action_repeat 4, contract v1`.
- **Census**: characterize_demo_sets.py reads contract-v1 tapes natively (stride table becomes exact-by-construction, reported anyway).

4.2 Adapters (inside record_demos.py)

- `human`: waypoint follower on the recorded absolute command stream of the human demo (rerecord_delta_demos.py logic generalized to delta_ref=target): per decision a_arm = clip((cmd_j − target_now)/(4·cap), −1, 1), a_grip = grip_j·2−1; advance j by arrival (‖q_meas − ref_j‖∞ < tol) up to 4 waypoints per decision, or dwell cap; time dilation counted; settle; kept iff hardened pick re-earned.
- `dp`: lerobot policy (dp_runner) queried once per decision on the current obs; output absolute target q* → a_arm = clip((q* − target_now)/(4·cap)); grip from the chunk; `--mode sample|mode`.
- `r2d`: champion loader lifted from harvest_champion_demos.py; native delta actions.
- `random`: uniform [-1,1]^7 per decision (negative control, must keep ≈0).

4.3 Terminal unification (offline)

`full_env.terminal_from_tape(tape) → (t_term, kind, reward)`: for contract-v1 tapes it READS `terminated/picked/tipped`; for legacy stride-1 tapes it computes the relabeler pick predicate + the tip rule and WARNS that the hardened eef distance is unavailable. Every legacy encoder/converter (train_sacfd_full delta encoders, to_dreamer_demos.py, convert_genesis_demos_repeat.py, make_dDPsucc tiptrunc) gets `--demo-terminal-guard` DEFAULT ON: tip → done=True r=0 and drop later frames; pick → done=True; cap → bootstrap. Unit tests on synthetic tapes: tipped tape ends done=True r=0; pick ends done=True; timeout ends done=False; contract-v1 tapes pass through unchanged.

5. Evaluation protocol

- **Harness**: `cluster/eval_sweep.sh` (NEW) drives `baselines/wandb_eval.py` ONE episode per fresh process, CPU, ≤1 world per 2 cores, `--max-steps 1200`, repeat/action-mode from the checkpoint sidecar (no silent defaults), for sac AND dp (dp gets the hold-4 path). r2d/dv3 evals go through their own eval scripts with the SAME IC file, horizon and fresh-process rule (patches shipped from here; cluster-side apply).

- **ICs**: `baselines/eval_ics.json` (NEW, generated once, committed): `sel` = 15 demo uids (selection), `hold` = 15 OTHER success uids (confirmation), `rnd` = 30 support ICs drawn once from rng(0). Same file for every learner/source/arm.
- **Checkpoints**: K=5 archived at 20/40/60/80/100% of budget for every run (RLPD callback; DP save_freq; r2d/dv3 archive step). Selection = best `sel` score; **headline per seed = the selected checkpoint on `hold` (15) + `rnd` (30)**; also report final-checkpoint on hold+rnd and time-to-first-training-pick.
- **Action selection**: RLPD deterministic; DP sampled (seeded); r2d sampled (+mode reported); dv3 deterministic — recorded in the result JSON. Denominators asserted; missing cells reported missing, never 0. Node class recorded per eval process.

6. Hypotheses (falsifiable; one primary statistic each)

Primary statistic = per-seed success count on `hold` (15) and on `rnd` (30) of the selected checkpoint; unit = seed; paired by seed index.

- **P-DP**: mean per-seed hold success differs between dH and each model set (2 contrasts, Holm). Prediction (from prior data, to be re-tested): dR2D > dDP > dH on hold; dR2D > {dDP, dH} on rnd. Falsified if the paired-difference CI covers 0. Epoch check: if the 300k-step view closes the dH gap, the 100k gap is attributed to under-training.
- **P-RLPD (sparse)**: success-only dDP ignites ≥ dH (N1); falsified if dDP remains below dH by ≥0.15 with CI excluding 0. **P-RLPD (dense)**: source ordering dR2D ≥ dH ≥ dDP on hold mean; dense raises every source's mean vs sparse (paired by seed).
- **P-R2D**: sparse — ignition lottery persists on all sources, dR2D pixel set ignites least (team's registered expectation); dense — 4/4-class ignition on dH replicates and extends to dDP/dR2D (falsifier: dR2D-dense ≥ dH-dense breaks "model pixel sets don't enter the basin").
- **P-MECH** (secondary): adding DP fails to dR2D lowers RLPD-sparse success by ≥0.15; removing them from dDP raises it to within 0.1 of dH.
- **P-DV3** (exploratory, last in queue): at 62.5k updates ≥1 seed per source sustains nonzero hold success at its last two archived checkpoints.
- **H4'** (exploratory): the SIGN of (model − human) differs between DP and r2dreamer; labelled exploratory because modality and budget differ across learners (CRITIQUE N2).

7. Analysis plan

Paired permutation test on seed means within learner; hierarchical Beta-Binomial P(A>B) (analysis/bayes_source_effect.py) as the reported posterior; 95% bootstrap CI on the mean difference as effect size. Two primary contrasts per learner → Holm; everything else BH q=0.1 and labelled exploratory. Ignition counts descriptive only. No pooling of episodes across seeds. Per cell report: n seeds, per-seed sel/hold/rnd, mean ± SD, CI, posterior, selected ckpt step, final-ckpt rate, time-to-first-pick, demo sha, N episodes/frames/rewarded frames, horizon/repeat/action mode, node class, video grid.

8. Pilots (positive controls at the new clock) — pass bars fixed now

```
pilot runs pass bar on fail
-------------------------------------------------- -----------
---------------------------------------------------------
------------------------------------------------------------
dH re-record at repeat 4 (recorder, human adapter) all 66 ≥50 survivors re-earn the hardened pick lower tol /
investigate; if <45, fallback: DP at stride 1 on
the same episodes (per-learner clock)
DP-r4 on dH 2 seeds ≥0.5 in-dist on sel (prior 0.62) DP stays stride 1 on the same episodes; amendment logged
RLPD-r4 dH dense 3 seeds pooled hold ≥0.16 and ≥1 seed ≥4/15 (08-19 dense verdict) keep repeat 1 for RLPD;
amendment
r2d dense dH at time_limit 1200 2 seeds ≥1 seed ignites by 1M with best-ckpt ≥0.5 keep 400; amendment
dp-adapter hold-4 harvest yield 30 rollouts ≥50% of the teacher's stride-1 yield switch teacher or stride;
amendment
random-teacher negctl 30 keeps ≈0 (≤1) recorder keep predicate broken — stop
```

9. Build list (what is being implemented now, from this doc)

B1 `baselines/record_demos.py` + adapters, contract v1, manifest, negctl. (genesis-side; runs on devbox/cluster) B2 `full_env.terminal_from_tape`; `--demo-terminal-guard` default-on in every legacy encoder/converter; φ(terminal)=0 in full_env.step; RLPD `--demo-format native` with demo shaping from eef_pos; unit tests on synthetic tapes. B3 `baselines/eval_ics.json` generator; wandb_eval `--ic-file` + DP hold-N; `cluster/eval_sweep.sh`; K=5 checkpoint archiving (train_rlpd callback, sbatch_dp save_freq); sbatch knobs (horizon, repeat, budget unit, demo sha), node class, CPU evals, registry at job start. B4 `baselines/make_matched_sets.py` (N=min, stratified by ic_uid, sha, manifest, per- learner derivations: npz/lerobot fps 7.5/dreamer-native) + `to_dreamer_native.py` + census reading contract v1; patch file for dreamerv3-torch-genesis (converter slack, eval IC file, time_limit, archive). B5 cluster-side TODO for the user (cannot be done from this box): pull 08-20 artifacts; verify r2dreamer discount in its config; r2d time_limit/archive/eval patches.

10. Disclosure paragraph (write now, spend no GPU)

State vs pixel modality; per-learner budgets and units; human 30 Hz tapes only ~half exactly representable at repeat 4 before re-recording (census) — after re-recording, representability is exact by construction for all sources and the human set's time dilation is reported; teacher circularity and competence; recipes tuned on dH; eval env has no tip termination; RLPD has no passing positive control; dv3 is last-in-queue and exploratory; single task, single simulator, N≈66; the 08-19 dDP_RLPD 0/6 is the motivation for the terminal guard and the recorder, not a result about model demos.

Amendments (appended, dated; never edited in place)

- **A1 (2026-08-23, pre-launch).** dH N = 51 of 66 (recorder: 15 human demos not executable inside the MDP — 2 lying-can ICs, 1 knock-over, 5 over-horizon, 7 over-cap; deterministic). Reported as a source property; no horizon stretch for the human adapter (§2 horizon held fixed).

- **A2 (2026-08-23, pre-launch, conditional).** dR2D teacher = r2dreamer round-robin dH-dense s51 `BEST_selected.pt` (cluster) if CHAMPION_1576820.pt is not available before the matched sets are built; same learner family, trained on the same human set, confirmed ~1.0 on sel. Logged in the set manifest (teacher_ckpt) and in the census.
- **A3 (2026-08-23).** `--ic-from-tape`: the 12 demos without a recovered placement reset to the can pose recorded at frame 0 of their own tape; applied identically to all three sources so every source records from the same 66 ICs.

11. Independent-audit schedule (locked by user 2026-08-23)

Milestone-gated, not calendar-driven. Each audit = one fresh-context agent (no conversation history; repo + docs only), read-only, must cite file:line, report archived under paper/AUDIT_*.

```
gate audit must answer
----------------------------------------------------- ------------------------------------------------------------
------------------------------------------------------------
before any block launches (after sets + census exist) Pre-launch: PREREG vs code vs manifests every sbatch
implements §2/§5 (clock, horizon, harness,
registry knobs); set shas/census match §3.2; no silent
default reappeared
first results readout of each learner family Results: numbers vs logs vs protocol headline = selected-on-
hold+rnd; denominators, missing
cells, node classes, seeds disjoint; no cross-learner
sentence without its caveat
before a claim enters the paper text Claim (adversarial): the claim, its evidence, its falsifier what a reviewer
rejects it on; cheapest experiment that
would kill it
after any trainer/env/encoder change mid-block Change (narrow): diff + tests does it restart that learner's block
(§2 "one commit hash")?
what else reads the changed path?
once, before submission Reproducibility: a stranger reruns one cell from repo + runbook, shas, registry, one cell
re-executed from scratch
rsynced sets
```

Next due: Pre-launch audit, 2026-08-24, after Phase 4 of the session runbook and before the DP/RLPD block.

- **A4 (2026-08-23, pre-launch audit B2).** Policy device: the DP policy runs on the GPU when one is visible (eval_sweep.sh dp path, recorder dp adapter, dDP recording jobs with GRES=gpu:1); the SIM is CPU always and SAC policies stay CPU. §5's "CPU" wording referred to the sim.
- **A5 (2026-08-23).** §8 pilots as actually run: random-teacher negctl n=3 (0 kept) not 30; the dp-adapter yield pilot was skipped — the full dDP harvest's first-attempt rate (0.55-1.0 per shard, 64/66 ICs) stands in; the RLPD-r4 dH dense pilot was NOT run separately — a 3k-step dense smoke (W1) plus the block's dH dense seeds 10-13 double as it; if dense RLPD at repeat 4 fails its bar the RLPD block restarts at repeat 1.
- **A6 (2026-08-23).** §4.2 human adapter arrival test: `--arrival either` (advance when the measured joints reach the human's measured pose OR the env's integrated target reaches the human's command; lab result paper/real2sim_follower_lab_2026-08-23.md §3: 49→55 local, zero dwell

stalls, same path fidelity). If its cluster confirmation keeps ≥55, the block's dH is the `either` set, built into a NEW root (matched_v2; matched_v1 is never rebuilt in place). The dDP teacher (DP-r4 pilot s0 ckpt 020000) was trained on the 51 `meas`-arrival tapes (sha 58fd89bc) — a superset/variant relation to the block's dH, disclosed.

- **A7 (2026-08-23).** Checkpoint selection ties → the LATER checkpoint (sbatch_rlpd.sh, dp_select_confirm.sh).
- **A8 (2026-08-23, audit W8).** `hold` is a CONFIRMATION split of demo ICs (11/15 are training ICs for a 51-demo set; for RLPD every sel/hold uid is a training-reset IC), not a held-out set; only `rnd` (30 support ICs) is novel. Report "sel / hold / rnd" by name; never call hold "held-out".

A9 (2026-08-28, from AUDIT_results_2026-08-28 §2) — run identity

A seed id names exactly one training run. A relaunch of an (arm, seed) that has already produced a readout REPLACES it: the earlier readouts are dropped from every pooled statistic and listed as lost. Relaunches must therefore use fresh seed ids; in-place reruns of completed seeds are forbidden. Retroactive: dR2DDPfails s83 (completed 08-26, sel 0.80) was overwritten 08-27 and is lost; dH s102's in-place relaunch was cancelled 08-28 and resubmitted as s108; dDP s101's relaunch never ran. `analysis/results_table.py` dedups on (arm, seed). Registry rows must carry a `warn` field when a semantic duplicate is accepted.

A10 (2026-08-28) — N15 controls (N16)

Two success-content controls bound (do not remove) the length/volume confound of the fails arm: dR2Ddup13 (share-matched) and dR2DDPsucc (8 longest dDP successes). Predictions in PAPER_NOTES N16.

A11 (2026-08-28) — ONE primary world for the learner x source matrix

The learner x source question (1a/1b) is answered in the OLD world (`base`, `matched_v2`): it is the only world in which all three sources exist for every learner, and every fails arm and every world-model run lives there. The corrected world (`gc_kp4_riser3_shelf6`, `matched_w3`) is a robustness replicate for dH/dDP under DP and RLPD, and the sim-fidelity work is reported as its own contribution. Re-teachering dR2D in the corrected world and re-running the matrix (~1 week of the 10-GPU cap) is out of scope for the deadline; the corrected-world sets are kept. Consequences applied 08-28: DP old-world top-up seeds 15-19 (n=5 -> 10 per source); RLPD old-world sparse top-up seeds 14-19 (n=4 -> 10); old-world split halves dH/dDP/dR2D _A/_B x seeds 50-52 (N14 in the primary world; the matched_w3 halves keep running as the replicate). Disclosure: the old world carries the three physics defects (no arm gravity compensation, buried shelf, soft contact) -- identical for every learner and source, and inert for the shelf in pick scope.

A12 (2026-08-28) — matrix simplified to human vs DP demonstrations

USER: "If DP does well, we can simplify to human vs DP demos until we're satisfied that everything is working correctly." DP does (rnd 0.5-0.6, hold ~0.9, stable across worlds). PRIMARY DESIGN from here: {dH, dDP} x {success-only, + the source's OWN fails} x {DP, RLPD, r2dreamer, dv3}, old world (A11). dR2D arms are RETAINED where already paid (old-world DP n=5, RLPD n=4, r2dreamer 80-91, N15/N16 mechanism cells) and reported as a secondary lineage note, not a matrix row. Cancelled 08-28 before start: DP dR2D top-ups 15-19 and dR2D split halves; RLPD dR2D top-up 14-19 and dR2DR2Dfails; r2dreamer sparse dR2D 80-83, dR2DR2Dfails 80-83, dR2Ddup13 200/201 (the packed pilot 202/203 kept as the packing test). Launched: r2dreamer dDPfails 80-83 (the WM 2x2 was missing its dDP+DPfails cell). The `dR2DR2Dfails` set (built, sha 248d0ade) is kept on disk for a later lineage appendix.

A13 (2026-08-28) — old vs corrected world: decide by the WM ignition gate, not in the abstract

USER: the ecological-validity question is old sim vs corrected sim. With dR2D dropped (A12) the corrected world (gc_kp4_riser3_shelf6, matched_w3) can carry the full 2x2x4: DP and RLPD dH/dDP success-only cells are already n=10 there. Missing and now queued: corrected same-source fails arms (matched_w3/dHHfails from dH_w2_fails, matched_w3/dDPfails from dDP_w2_fails) for RLPD sparse x seeds 20-23 (8), and the WM IGNITION GATE r2dreamer dense dH/dDP x seeds 80-83 on matched_w3 native tapes, TL 400 (8). RULE: if the gate ignites (>= 2/4 seeds per arm above 0.3 on the protocol readouts), the corrected world becomes the PRIMARY world for the matrix on validity grounds and the WM fails arms are run there too; the old world becomes the replicate. If it does not ignite, that is reported as a result and A11 stands. Known cost either way: RLPD sits near the floor (~0.2) in the corrected world, so the old world remains the sensitive readout for RLPD's source contrast.

A14 (2026-08-28) — learner health before source comparisons

USER: "the top priority is ensuring RLPD and WM runs are behaving appropriately before we worry about human vs machine demos." Queue reprioritised: all human-vs-machine power jobs HELD (35 DP top-ups/halves/N7, 29 r2d seed top-ups/sparse/fails arms, 24 old-recipe RLPD); runnable = the diagnostics only (18 RLPD fix factorial N18, 9 dv3 baseline+clamp/fix, 4 dv3 touchgoal, 2 r2dreamer touchgoal, 8 corrected-world WM gate, 1 packing pilot). A health audit of r2dreamer (critic pinned at the clamp? checkpoint bistability? selection bias?) runs in parallel. Held jobs are released only after each learner's recipe is judged healthy; if a recipe changes, the held jobs are cancelled and resubmitted under the new recipe.

A15 (2026-08-28) — validity rule

USER: "Don't do anything that would cause our implementations to diverge significantly from the standard versions (we want validity most of all)." Applied: (i)

RLPD fixes may only move the recipe TOWARD Ball et al. 2023 (gamma 0.99 is the paper's value; our 0.998 was the deviation); (ii) r2dreamer's return clamp and replay-critic loss are DEVIATIONS from DreamerV3 -- if the health audit finds them load-bearing, a standard arm (clamp off, repval off) is run and reported, not hidden; (iii) the dv3 `return_clamp` overlay is diagnostic only; the by-the-book dv3 arm is `genesis_dv3std` (reward_scale 1 with symlog/two-hot heads, no clamp, stock train_ratio) on terminal-reward-1 native demos -- launched dH dense x seeds 20-22.

A16 (2026-08-29 06:30, BEFORE any g99 s33-37 readout) — RLPD source statistic

The RLPD human-vs-DP contrast is scored on (i) **divergence rate** per arm (a run diverges if max critic_loss ≥ 1 or mean actor-state Q exceeds the attainable return on a sparse arm at any logged step after 30k; Fisher exact, two-sided) and (ii) **rnd-30 success of the LAST checkpoint** (not the selected one; Welch + exact permutation as in analysis/stats.py). hold is reported but is not the statistic: dH sits at its ceiling (14/15 in 6/6 runs) and hold ICs overlap training ICs (A8). The Q-watchdog trip count is NOT a health criterion on dense arms (potential shaping legitimately lifts Q above 2.0; train_rlpd.py:274-276 rescales the threshold only for hold reward); dense arms use max critic_loss and final≈selected only. Predictions: divergence dH 0-1/8 vs dDP ≥3/8; LAST-ckpt rnd dH > dDP by ≥0.10.

A17 (2026-08-29) — RLPD block restarted at γ = 0.99

PREREG §2 listed γ 0.998 (a deviation from Ball et al. 2023). Every γ 0.998 run (seeds 10-19, 24-29, waves final/w2final) is superseded and diagnostic-only (E9). The block is re-run at γ 0.99, UTD 10, seeds 30-37 (fresh ids, A9). Dense arms relabel demo rows with the run's γ at load (train_rlpd.py:311), so γ 0.998 and γ 0.99 dense runs may never share a table. Same-source row-matched controls added: dHsucc_dup / dDPsucc_dup (56 successes + the 8 longest own successes duplicated, 6xxxxx stems; added rows 1755 / 1521 vs the fails arms' 2145 fail rows), RLPD sparse seeds 30-33. Reading rule for (1b): the fails effect is claimed only if +fails differs from +succ_dup in the same direction on both sources.

A18 (2026-08-29) — world-model fallback

If neither dv3 (standard recipe: genesis_dv3std) nor r2dreamer NOCLAMP (return clamp off, replay critic loss kept — the official DreamerV3 v2 component) shows a sustained pick with a bounded value estimate by ~1M sim steps, the WM arm becomes MoDem as published (via the DEMO3 codebase), including its BC-initialised actor; the actor-BC confound is disclosed and H4 is reworded to "world-model learners that back up value through demonstration trajectories". First dv3-std picks were logged at 284k (s20) on 08-29 05:40 — before this amendment; the 1M/3M readouts decide.

A19 (2026-08-29) — one world per learner table

A13's corrected-world WM gate is not load-bearing: the WM arm reports in whichever world it is healthy in (old world, where every WM run to date lives), stated in the table header; no table mixes worlds. The DP corrected-world result and the RLPD/WM old-world results are separate tables with the world named.